
بِسْمِ اللَّهِ الرَّحْمَٰنِ الرَّحِيمِ

Geometric Corrections

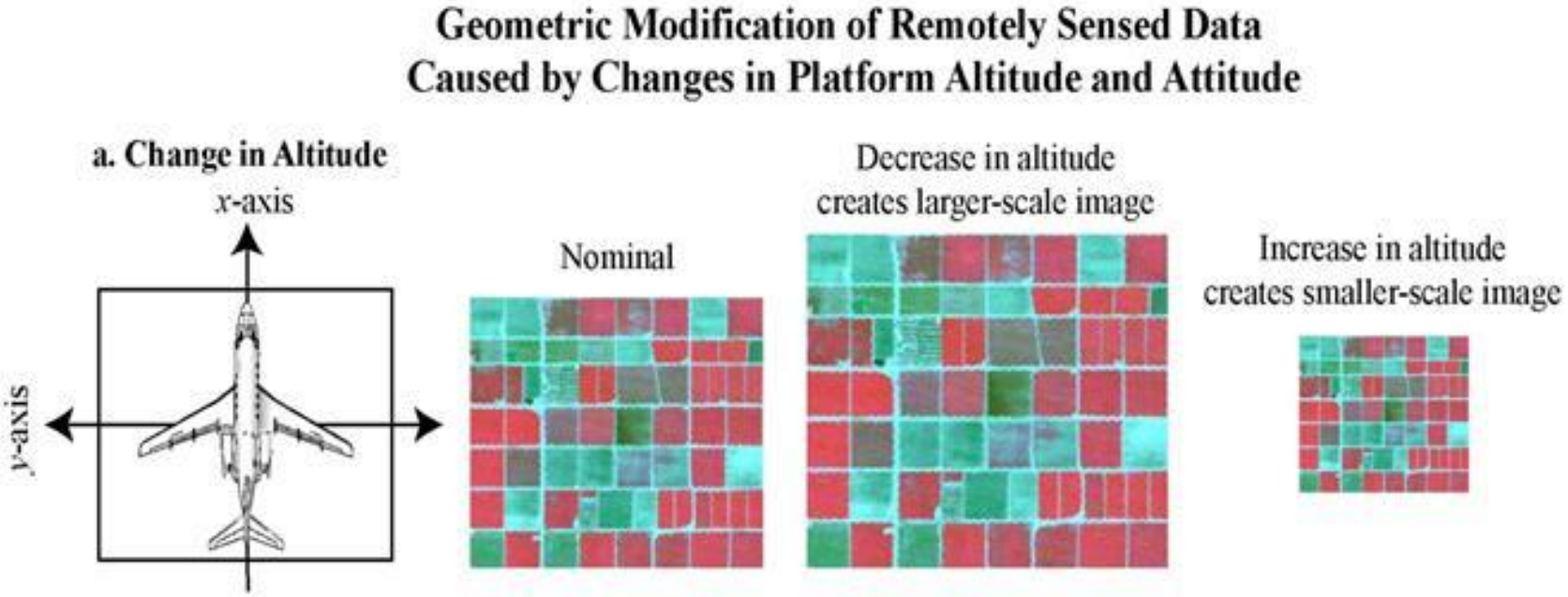
Un-Systematic (External)Geometric Error

- **The most important external variables that can cause geometric error in remote sensor data are random movements by the aircraft (or spacecraft) at the exact time of data collection, which usually involve:**
  - **Altitude changes, and/or**
  - **Attitude changes (roll, pitch, and yaw).**

Altitude Changes

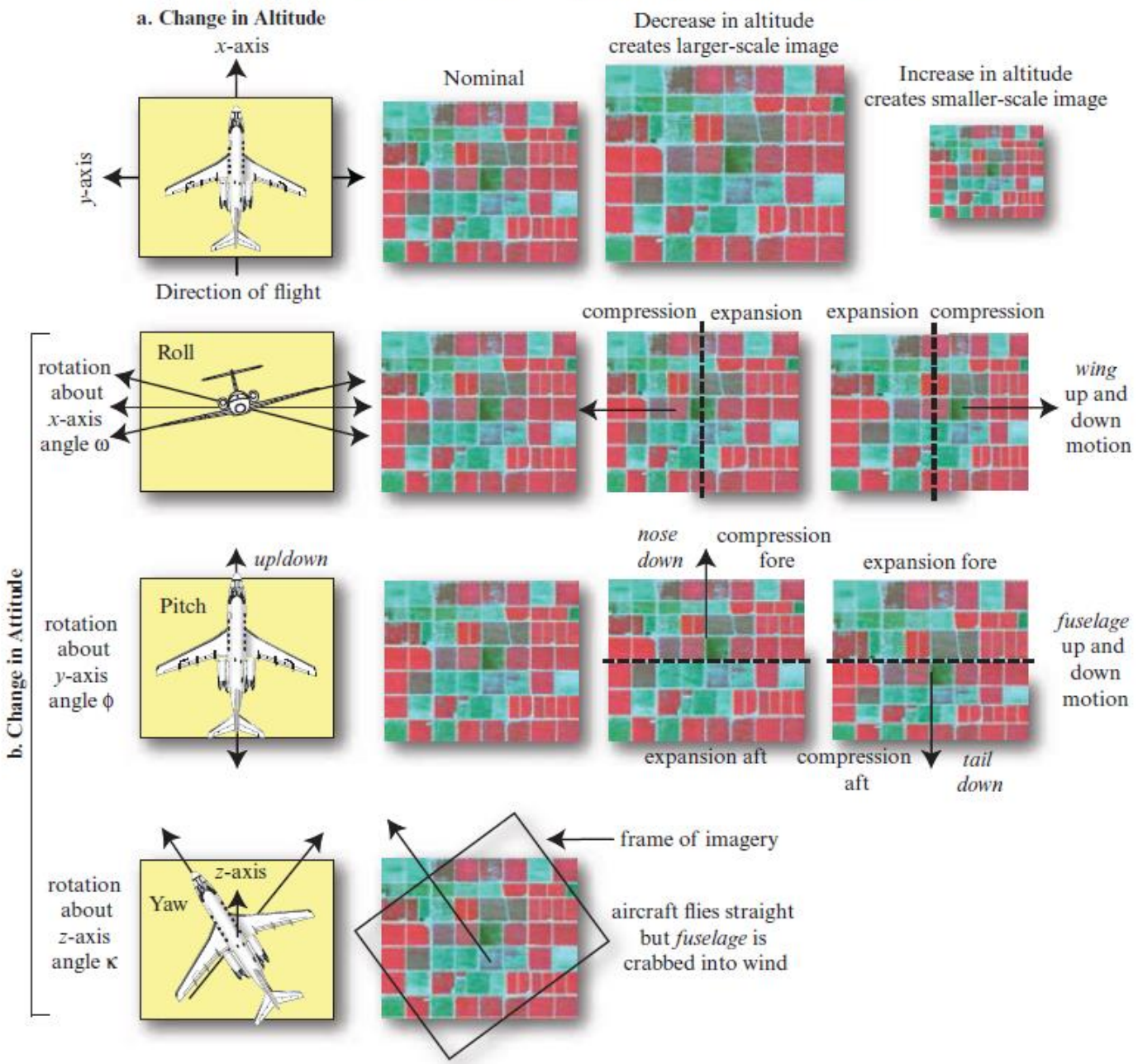
- Remote sensing systems flown at a constant altitude above ground level (AGL) result in imagery with a uniform scale all along the flight line
- If the aircraft or spacecraft gradually changes its altitude along a flight line, then the scale of the imagery will change
- Increasing the altitude will result in smaller-scale imagery (show large areas of land on a small space)
- Decreasing the altitude of the sensor system will result in larger-scale imagery (show smaller areas in more detail)
- The same relationship holds true for digital remote sensing systems collecting imagery on a pixel-by-pixel basis

Altitude Changes



Attitude Changes

- **Satellite platforms are usually stable**
- **Aircraft must constantly contend with atmospheric updrafts, downdrafts, head-winds, tail-winds, and cross-winds when collecting data**
- **Even when the remote sensing platform maintains a constant altitude AGL, it may rotate randomly about three separate axes that are commonly referred to as *roll*, *pitch*, and *yaw***
- **Quality remote sensing systems often have gyro-stabilization equipment that isolates the sensor system from the roll and pitch movements of the aircraft.**
- **Systems without stabilization equipment introduce some geometric error into the remote sensing dataset through variations in roll, pitch, and yaw that can only be corrected using *ground control points*.**


Geometric Modification of Remotely Sensed Data
Caused by Changes in Platform Altitude and Attitude
a. Change in Altitude
x-axis
y-axis
Direction of flight
Nominal
Decrease in altitude
creates larger-scale image
Increase in altitude
creates smaller-scale image
b. Change in Attitude
rotation
about
x-axis
angle ω
Roll
compression
expansion
expansion
compression
wing
up and
down
motion
rotation
about
y-axis
angle ϕ
Pitch
up/down
nose
down
compression
fore
expansion fore
expansion aft
compression
aft
tail
down
fuselage
up and
down
motion
rotation
about
z-axis
angle κ
Yaw
z-axis
frame of imagery
aircraft flies straight
but fuselage is
crabbed into wind

Un-Systematic (External)Geometric Correction

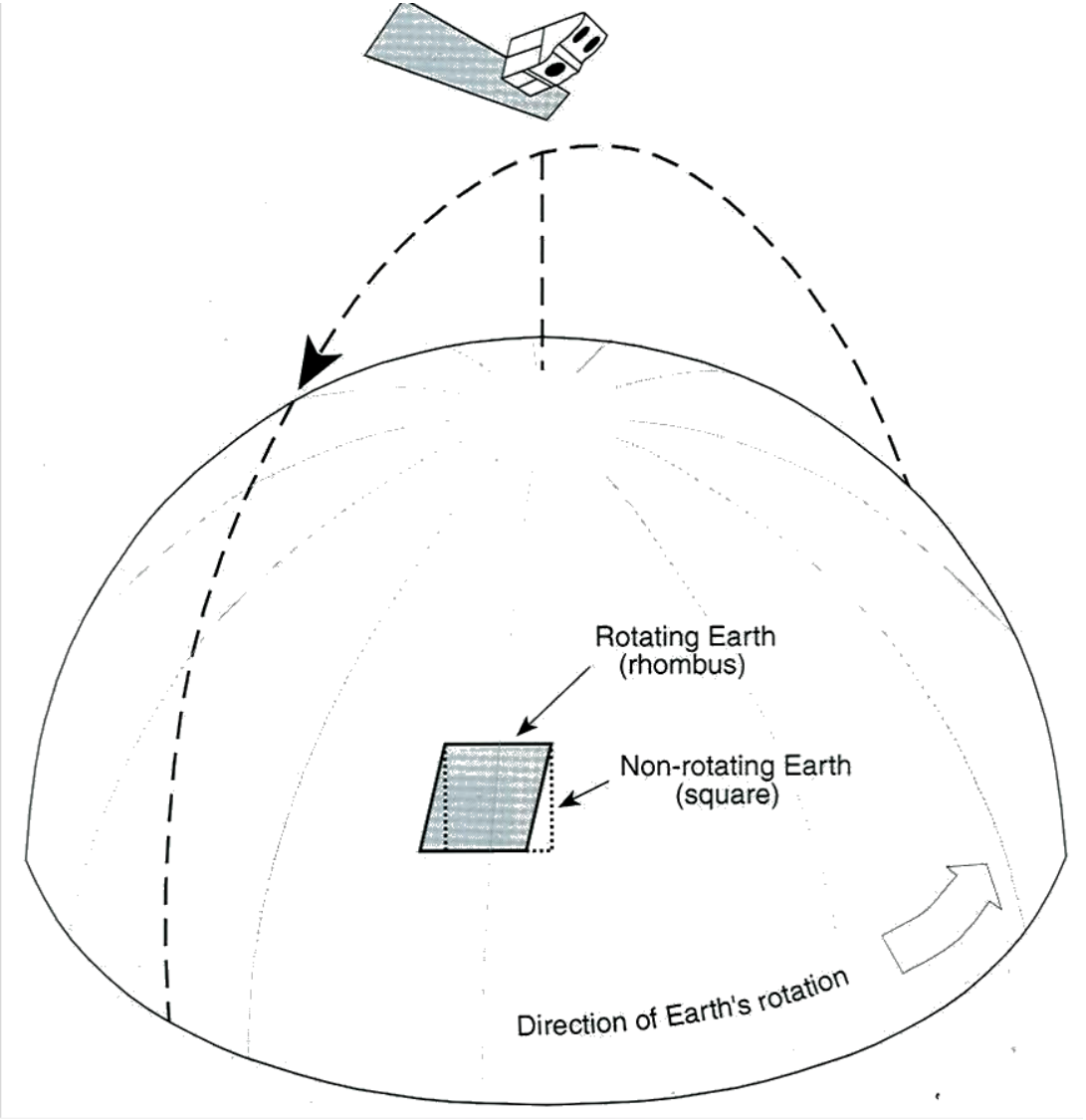
- **How will you treat and perform to rectify these errors?**

Systematic Distortions- Image Skew

Image Offset (*skew*) caused by Earth Rotation Effects

- **Earth-observing Sun-synchronous satellites are normally in fixed orbits that collect imagery while in the north to the south in descending mode**
- **Meanwhile, the Earth below rotates on its axis from West to East**
- **This interaction between the fixed orbital path of the remote sensing system and the Earth's rotation on its axis *skews* the geometry of the collected imagery**

Image Offset (*skew*) caused by Earth Rotation Effects



Systematic (Predictable)

- **Skew distortion –Earth Rotation**
  - **Cause the successive scan lines to start slightly further west than the previous scan lines, i.e.**
    - **Each optical sweep of the scanner covers an area slightly to the west of the previous sweep**
    - **Offset each successive scan line slightly to the west: parallelogram image**

Scan Skew

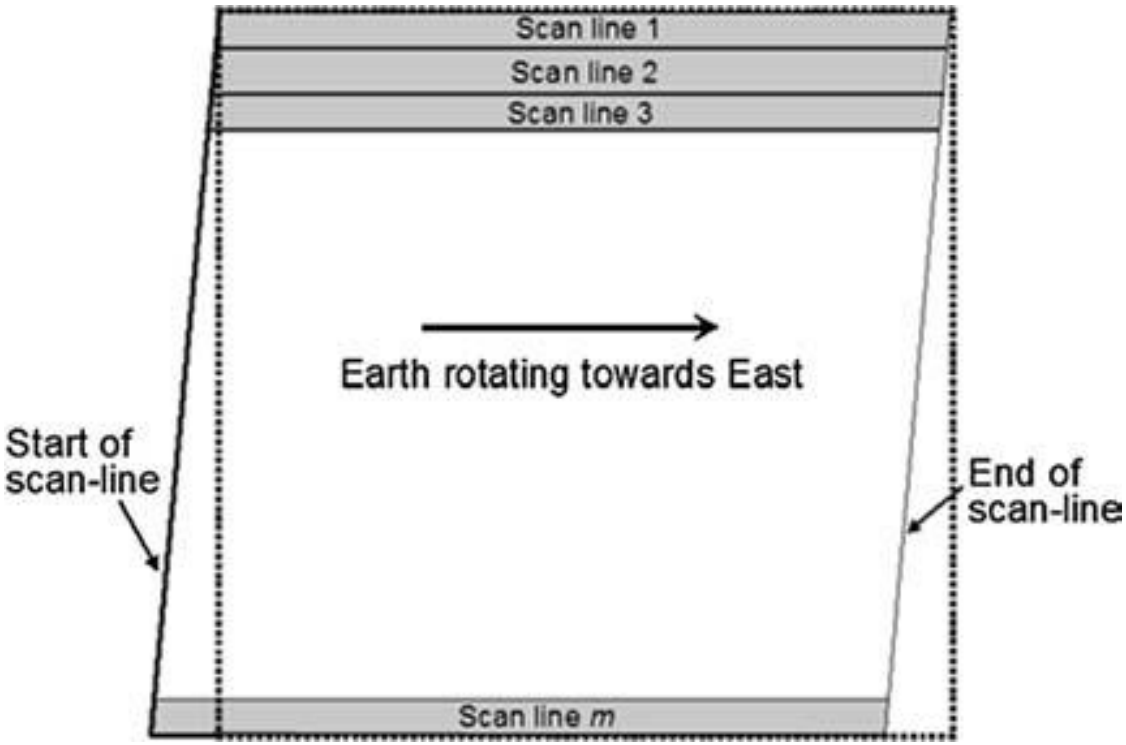
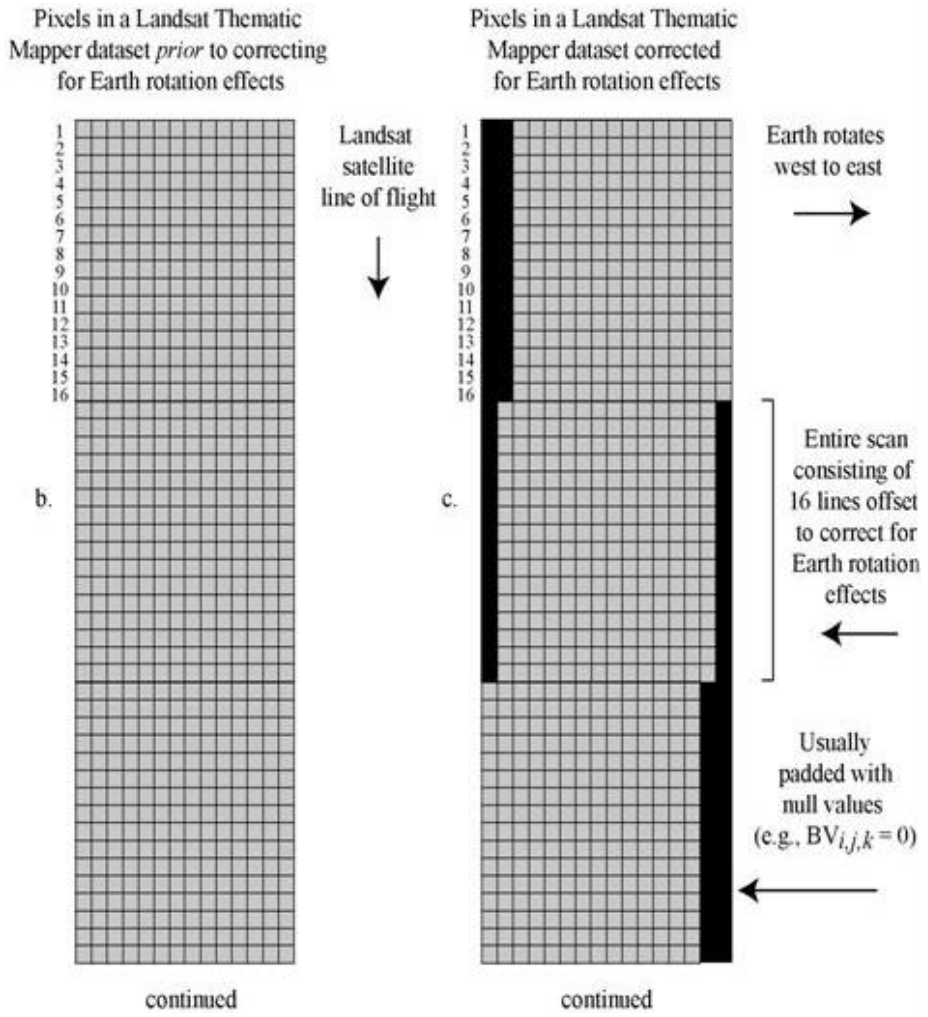
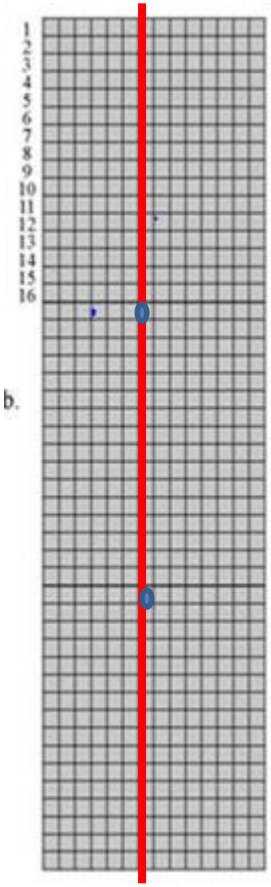

Scan line 1
Scan line 2
Scan line 3
Earth rotating towards East
Start of scan-line
End of scan-line
Scan line *m*

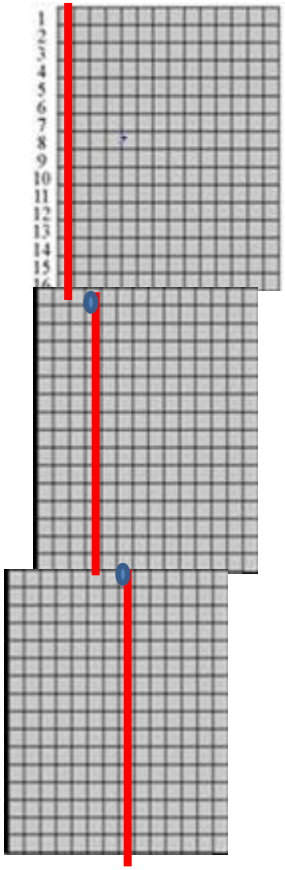
Image Skew-Deskew



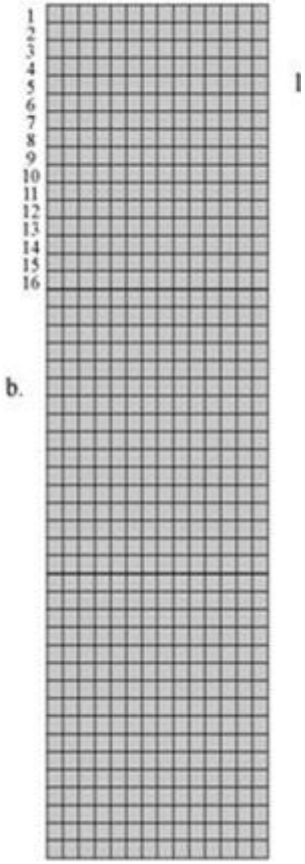
- **Consider a satellite as it moves southwards above the Earth's surface**
- **At time *t, its sensor scans image lines (1–16) and***
- ***At time t + 1,* lines 17–32 are scanned**
- **But the Earth has moved eastwards during the period between time *t and time t + 1***
- ***Therefore,* the start of scan lines 17–32 is slightly further west than the start of scan lines 1–16.**
- **Similarly, the start of scan lines 33–48 is slightly further west than the start of scan lines 17– 32**

Scan Line Acquisition and Skew





Pixels in a Landsat Thematic Mapper dataset *prior* to correcting for Earth rotation effects



continued

Skew distortion –Earth Rotation

- **Deskewing**
  - **Each successive scanner sweep is shifted slightly to the west**
  - **The skewed parallelogram appearance of satellite multispectral scanner data is result of this correction**

Deskewing

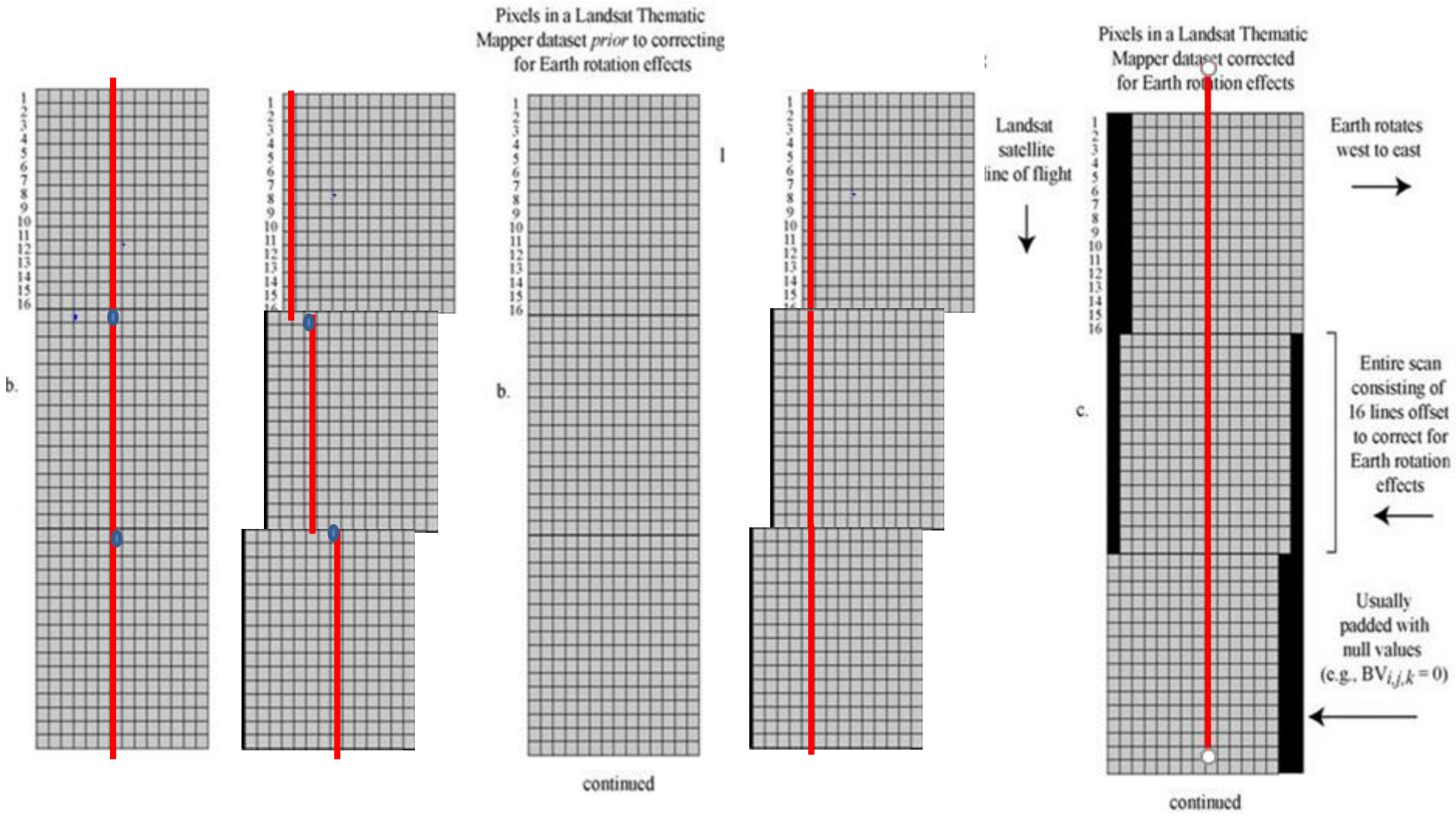
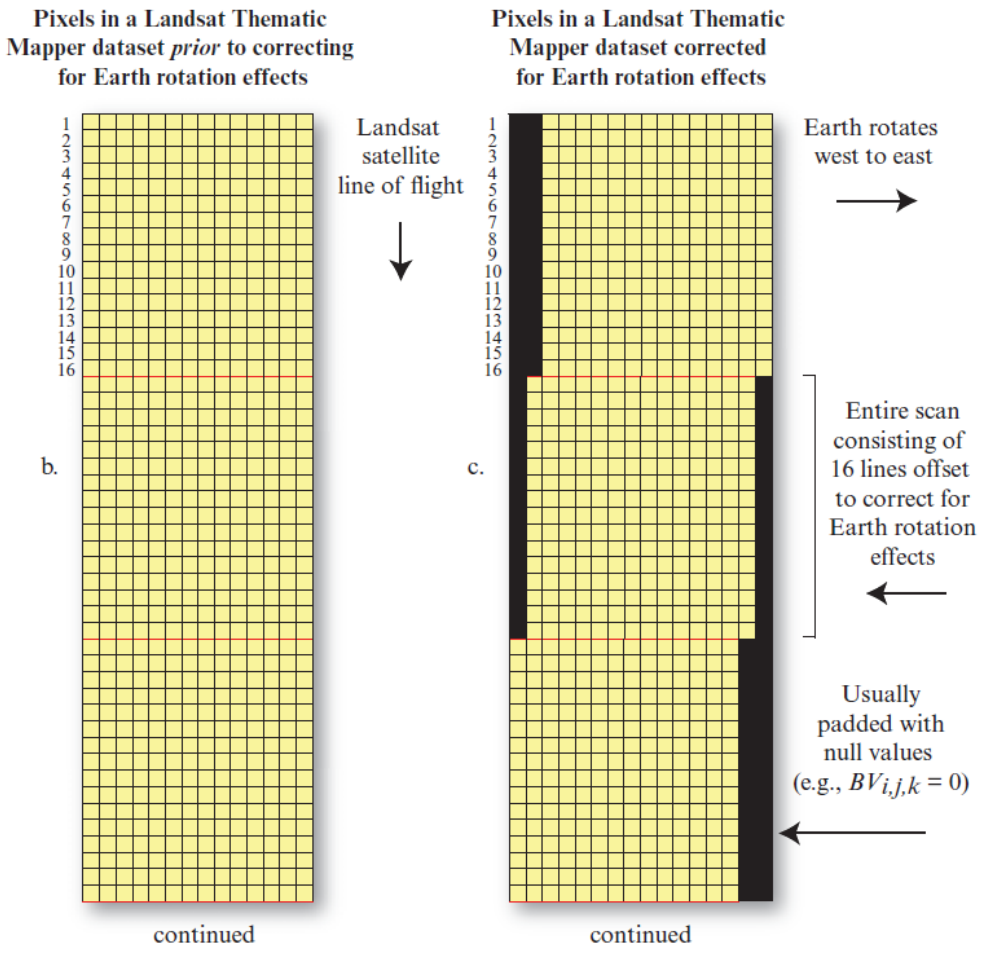
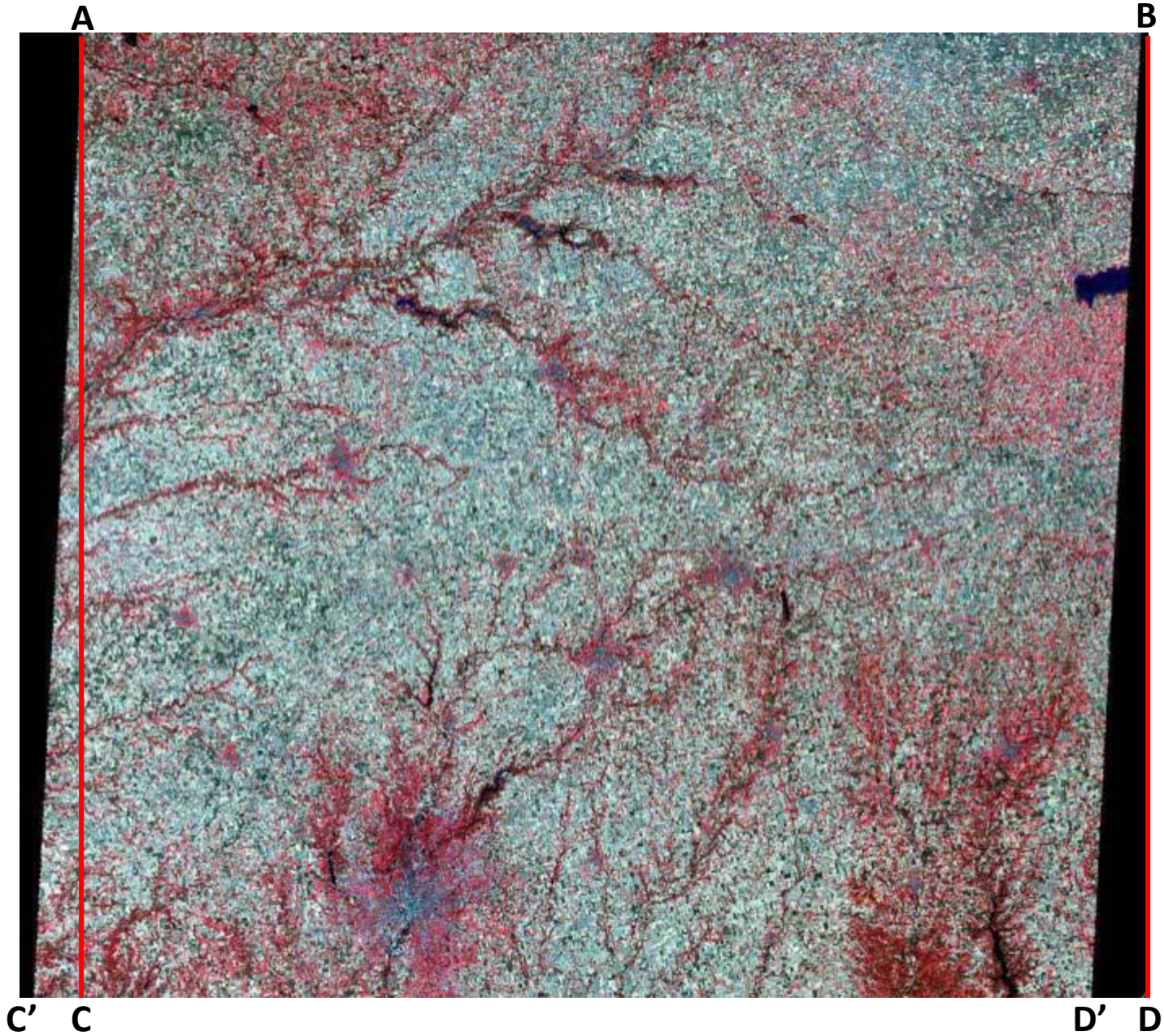

Pixels in a Landsat Thematic Mapper dataset *prior* to correcting for Earth rotation effects
b.
b.
continued
Landsat satellite line of flight
Pixels in a Landsat Thematic Mapper dataset corrected for Earth rotation effects
c.
Earth rotates west to east
Entire scan consisting of 16 lines offset to correct for Earth rotation effects
Usually padded with null values (e.g., $BV_{i,j,k} = 0$)
continued

Image Skew-Deskew



Skew distortion –Earth Rotation



Skew distortion –Earth Rotation

- **Deskewing**
  - **The amount by which the image has to be skewed to the west at the end of the frame depends upon**
    - **The relative velocities of the satellite and Earth and**
    - **The length of the image frame recorded**

Skew distortion –Assessment

- **Angular velocity of the satellite (ω_0)= mrad s−1**
- **L = frame on the ground (km)**
- **Time to scan this frame $t_s = L/(r_e\omega_0)$ = seconds**
  - **where r_e is the radius of the earth (637816 Km)**

Skew distortion –Assessment

- **Velocity (surface) of the Earth = $v_e = \omega_e r_e \cos \lambda$**
  - **where λ is latitude and**
  - **ω_e is the Earth rotational velocity of 72.72µrad s−1**

Skew distortion –Assessment

- During the frame acquisition time, the surface of the earth moves to the east by
  - $\delta x_e = v_e t_s$ = km
- Skew distortion:
  - % age of the frame size = Frame size/ δxe
- Since the satellite does not pass directly north south, this movement has to be corrected by the inclination angle
- So, the effective sideways movement of the earth is
  - $\delta x = \delta x_e \cos$ inclination ◦ = km

Panoramic Distortion

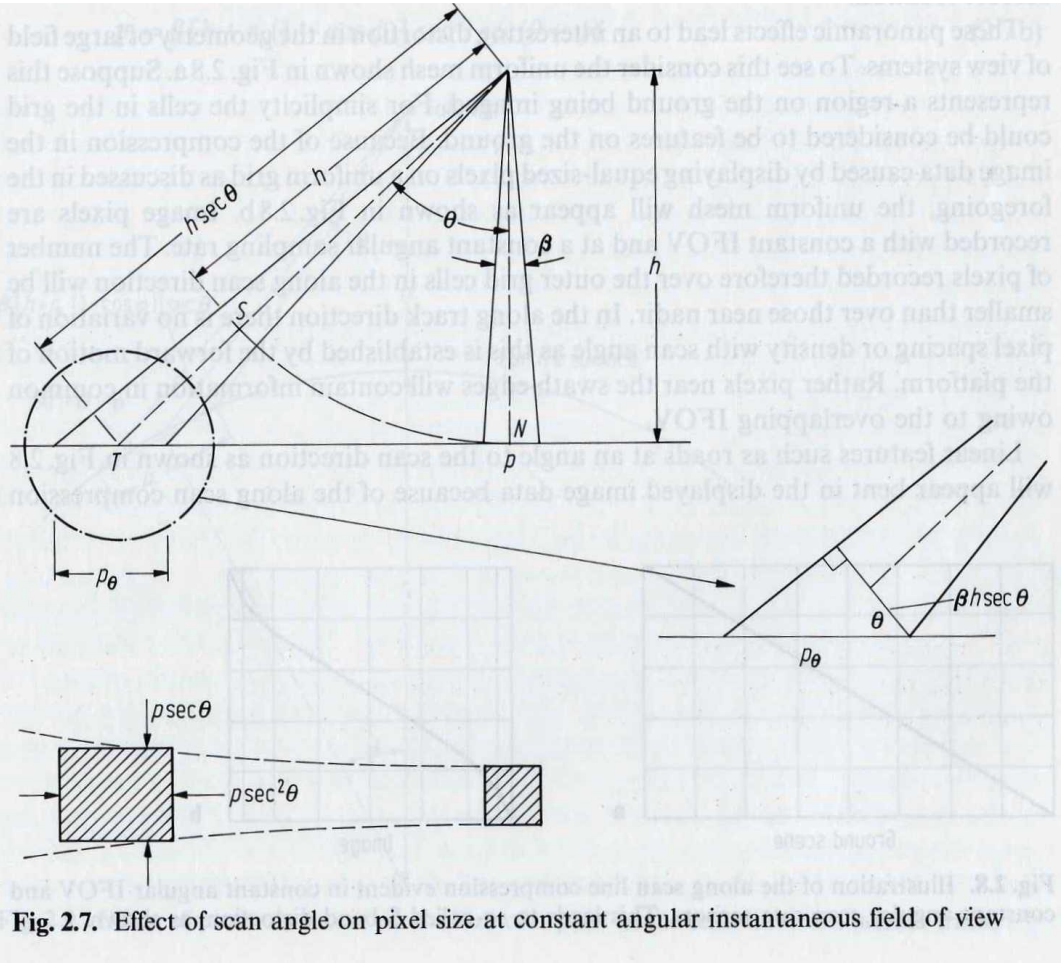


Fig. 2.7. Effect of scan angle on pixel size at constant angular instantaneous field of view

For scanners used on spacecraft and aircraft remote sensing platforms the angular IFOV is constant

As a result, the effective pixel size on the ground is larger at the extremities of the scan than at nadir

At IFOV β, the pixel dimension at nadir is p

Then the dimension in the scan direction (along) at a scan angle of θ as shown is

$$p\theta = \beta h \sec^2\theta = p \sec^2\theta$$

- **dimension across the scan line is *p* sec θ**
- **For small values of θ these effects are negligible**

Earth Curvature

$$p_c = \beta[h + r_e(1 - \cos\phi)] \sec\theta \sec(\theta + \phi) \qquad (2.7b)$$

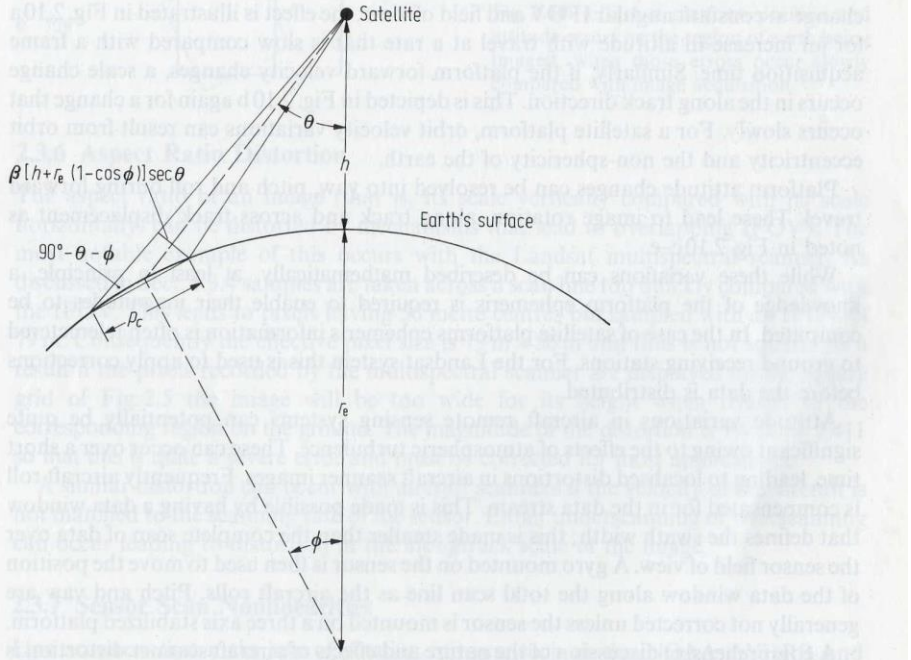


Fig. 2.9. Effect of earth curvature on the size of a pixel in the scan direction (across track)

Less swaths are not affected by the Earth curvature, AC, Landsat, SPOT etc, but Wide swath such as NOAA AVHRR are

Earth Curvature

- **Earth Curvature**
  - **Less swaths are not affected by the Earth curvature, AC, Landsat, SPOT etc, but**
  - **Wide swath such as NOAA AVHRR are**

Panoramic Distortion/Earth Curvature

- However, for systems with larger fields of view, the effect can be quite severe
- These panoramic effects lead to distortion in the geometry of large field of view systems
- Less swaths are not affected by the Earth curvature, AC, Landsat, SPOT etc, but
  - Wide swath such as NOAA AVHRR are

Relief Displacement

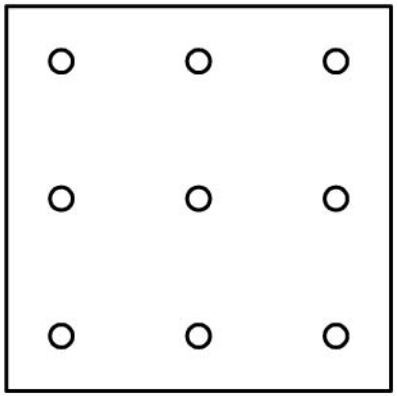
Relief Displacement

- **Shift in photographic position of image caused by topographic variations**
- **On a map a vertical object (power pole, building corner, etc.) has only one position**
- **On an aerial photo/ image it is often possible to see both the bottom and top of vertical objects, as if different points on a vertical object have different horizontal coordinates**
- **It is very apparent on vertical objects unless the object is directly below the center of the photo**

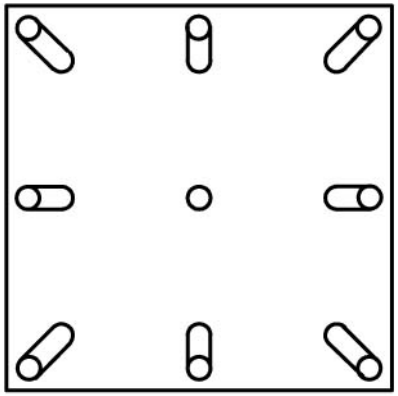
Relief Displacement

- Vertical object on photo appear to be lying along radial line to the nadir point
- Magnitude of the displacement in the image between top and bottom of the object is its relief displacement
- Is related to the height and its distance form the nadir point
- A vertical object at nadir will not be displaced, but the one at the edge of the vertical photo will appear leaning away from the nadir point
- Relationship between the object height and relief displacement can be derived using geometry of the vertical photo

Relief Displacement

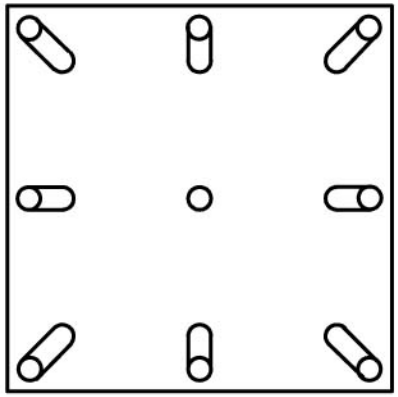


Object as visible on a map

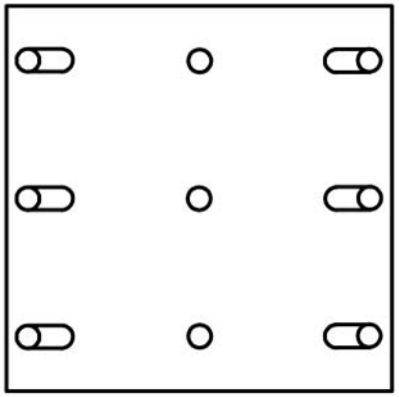


In frame imagery (entire image captured at same instant), relief displacement is always radial with respect to the nadir point

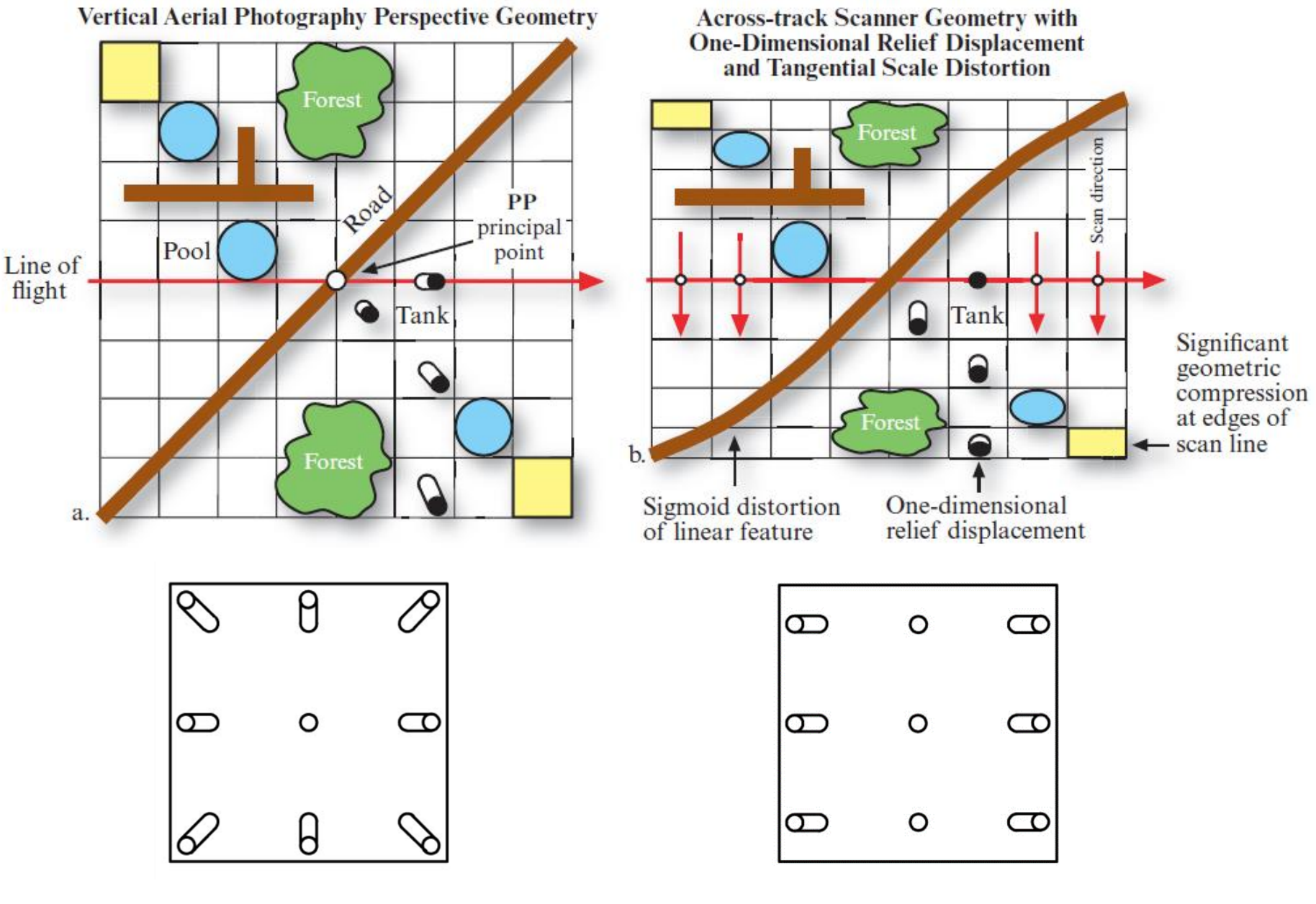
Scanning System One-Dimensional Relief Displacement



- Images acquired using along and across-track scanning system introduces one-dimensional relief displacement perpendicular to the line of flight
- However, instead of being radial from a single principal point, the displacement takes place in a direction that is perpendicular to the line of flight for each scan line

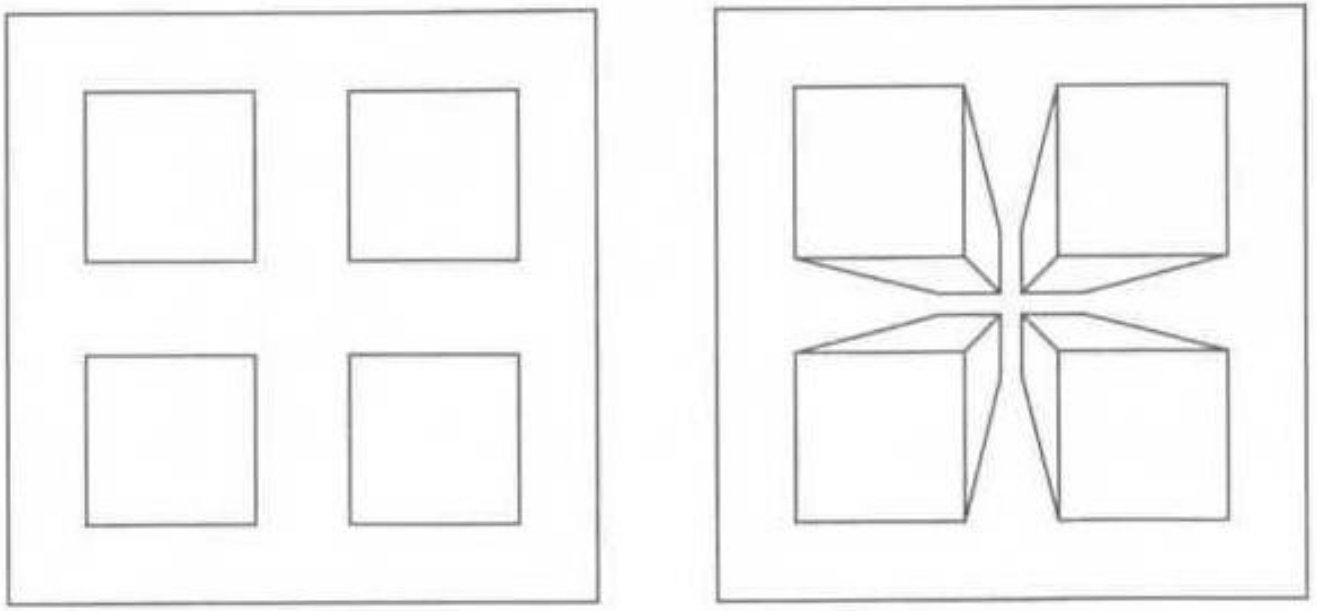


- In effect, the ground-resolution element at nadir functions like a principal point for each scan line
- At nadir, the scanning system looks directly down (e.g. on a water tank, and it appears as a perfect circle)

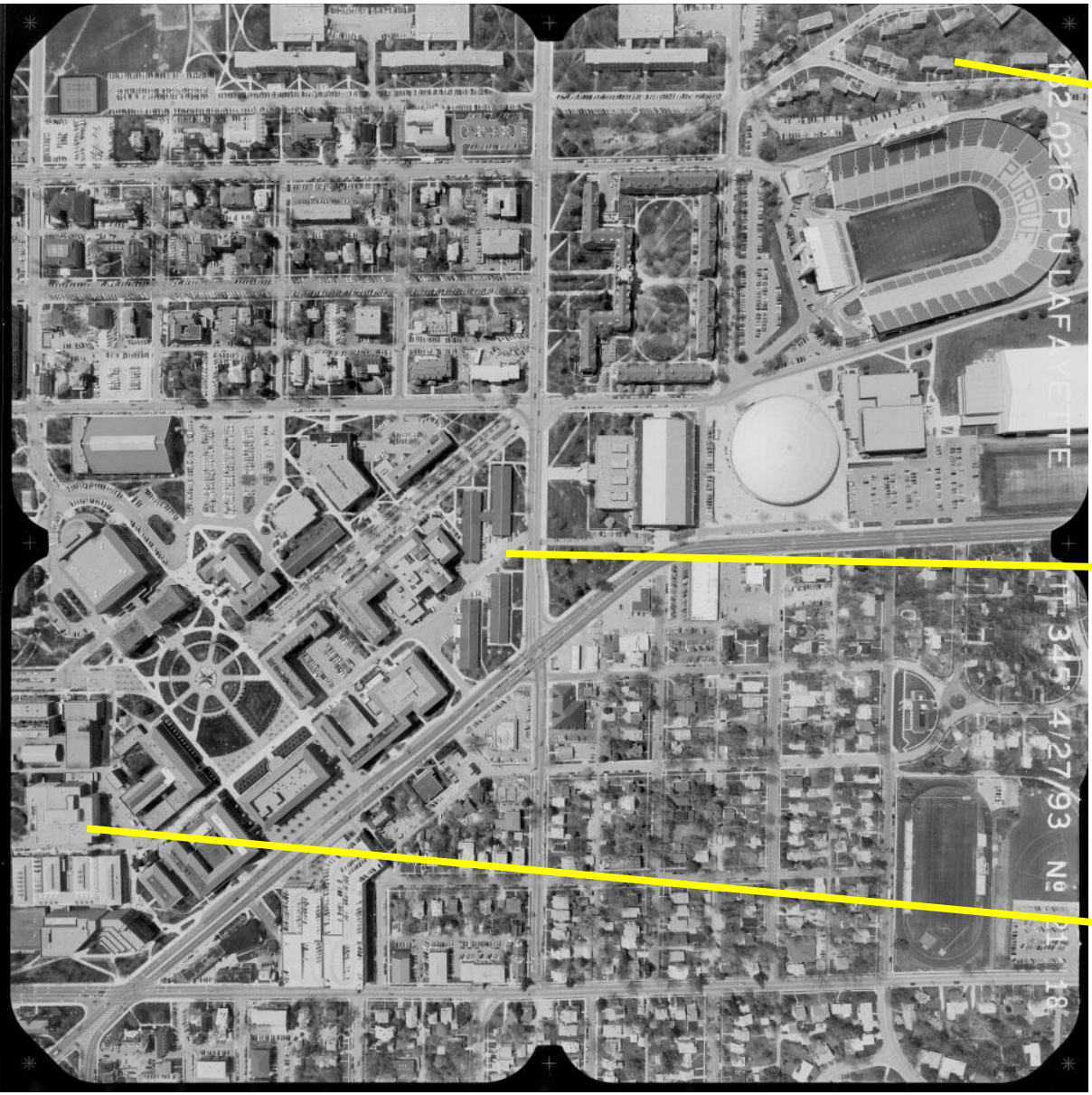

Vertical Aerial Photography Perspective Geometry
Forest
Road
PP
principal
point
Pool
Line of
flight
Tank
Forest
a.
Across-track Scanner Geometry with
One-Dimensional Relief Displacement
and Tangential Scale Distortion
Forest
Scan direction
Tank
Significant
geometric
compression
at edges of
scan line
Forest
b.
Sigmoid distortion
of linear feature
One-dimensional
relief displacement

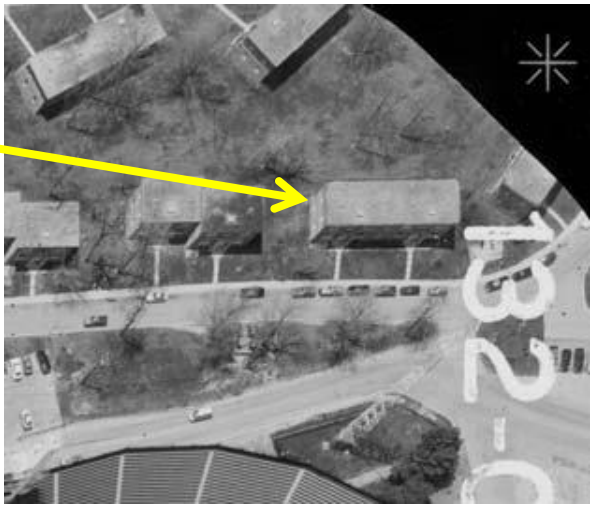
Orthographic and Perspective projection

- **Terrain relief causes an image point to be displaced from its planimetrically correct position**

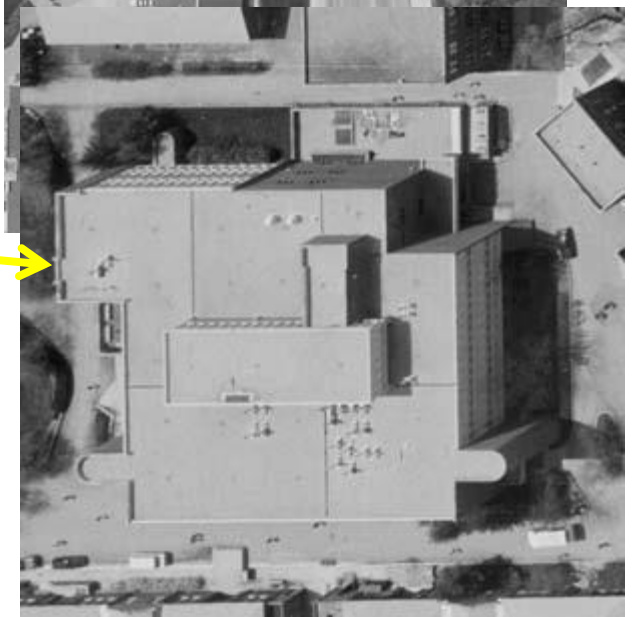


(a) (b)









Relief Displacement


5-25-73
24M-Q5
PRINCIPAL

Relief Displacement





Relief displacement in images taken at different locations

Ortho-rectification



Relief Displacement

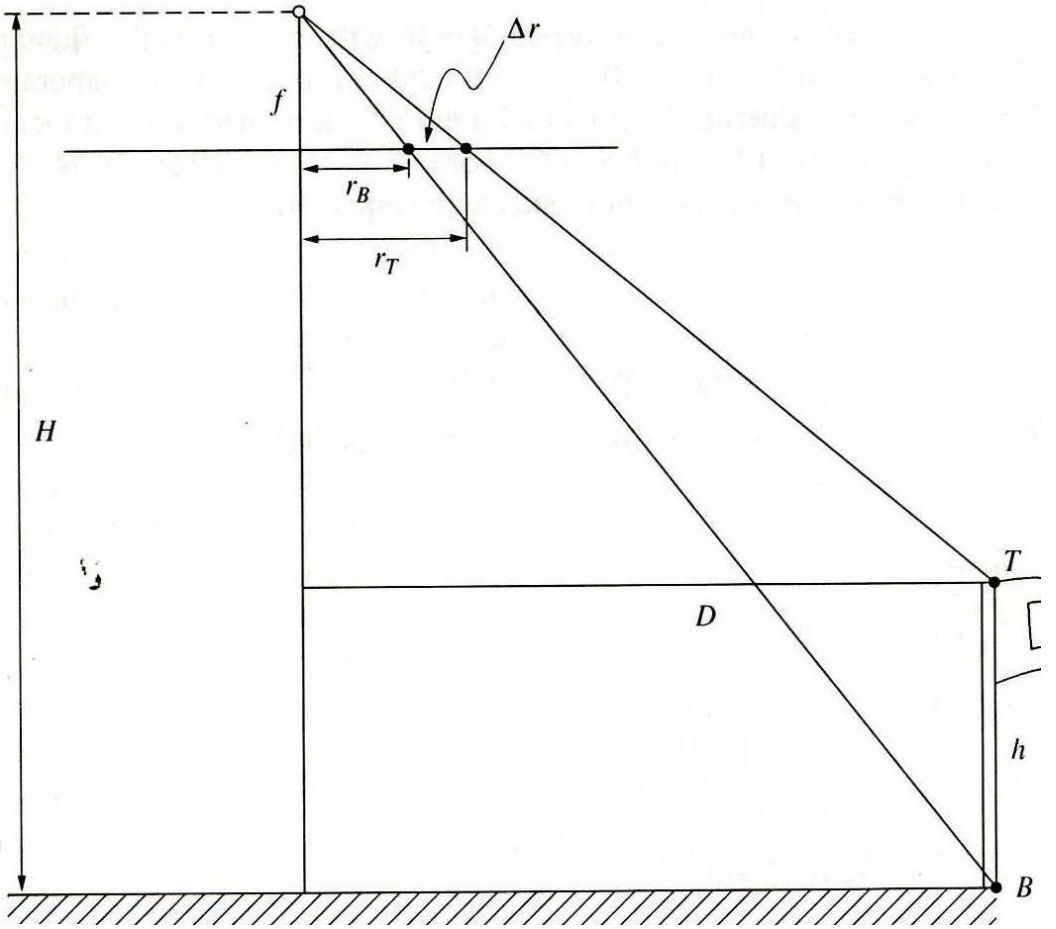
$$\frac{r_B}{D} = \frac{f}{H}, \quad D = \frac{Hr_B}{f}$$

$$\frac{r_T}{D} = \frac{f}{H-h}, \quad D = \frac{r_T(H-h)}{f}$$

$$D = \frac{Hr_B}{f} = \frac{r_T(H-h)}{f}$$

$$Hr_T - hr_T - Hr_B = 0$$

$$H(r_T - r_B) = hr_T$$



Relief Displacement

$$\Delta r = \frac{h\, r_T}{H}$$

Height of the Object $\quad \frac{H\Delta r}{r_T} = h$

f:focal length, H= Altitude

Relief Displacement

- **Amount of relief increases the farther the point is with respect to the principal point**
- **Higher the elevation, the more the amount of relief displacement**
- **The greater the height of the object above the local terrain, the greater the amount of *relief displacement***
- **Increasing flying height decreases the amount of relief displacement**